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Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

NAMMA RAITHA:A Mobile App Application For Farmers

AN INTERNSHIP REPORT

Submitted by

THANUSH G GOWDA – 20221COM0088

Under the guidance of,

Dr. Murali Parameswaran

BACHELOR OF TECHNOLOGY

IN

COMPUTER ENGINEERING
(Artificial Intelligence and Machine Learning)

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "NAMMA RAITHA:A Mobile App Application For Farmers" is a Bonafide work of **THANUSH G GOWDA (20221COM0088)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER ENGINEERING,ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING during 2025-2026.

29/4/26		24/04/2026.
Dr. Murali Parameswaran Professor Internship Guide PSCS Presidency University	**Mr. Sreehari TM** Associate Professor Program Internship Coordinator PSCS Presidency University	**Dr. Md Ziaur Rahaman** School Internship Coordinator PSCS Presidency University

24/4/26	
Dr. GopalKrishna Shyam Professor & Head of the Department PSCS Presidency University	**Dr. Duraipandian N** Dean PSCS Presidency University

Sl.No	Name of the Examiner	Signature
1.	Dr. Vishnu	29/4/26
2.	Dr. Vishwanath Y	

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DECLARATION

I am a student of final year B. Tech in COMPUTER ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING), at Presidency University, Bengaluru, named **Thanush G Gowda**, hereby declare that the internship work titled "NAMMA RAITHA:A Mobile App Application For Farmers" has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B. Tech in COMPUTER ENGINEERING (Artificial Intelligence And Machine Learning) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Thanush G Gowda 20221COM0088

PLACE: BENGALURU

DATE:

INTERNSHIP COMPLETION CERTIFICATE

Robowaves

CIN: U72200KA2007PTC044701

Date: 04/12/2025

TO WHOM SO EVER IT MAY CONCERN

This is to certify that **Mr/Ms. Thanush G Gowda** bearing **USN – "20221COM0088"** a student of **"Bachelor of Technology"** from **Presidency University, Itgalpura, Rajanukunte, Yelahanka.** had worked as intern and completed his/her internship in our company and worked on **"Web Technology"** from **01/01/2026 to 31/03/2026**. Successfully completed **3 months** of internship under the guidance of **Mr. Pradeep (Robowaves (A Unit of Test Yantra Software Solutions (India) Pvt. Ltd.)**

During this period of this internship program with us we found he/she is sincere and hardworking.

We wish success for all his/her future endeavours.

Yours Sincerely,

For Robowaves
(A Unit of Test Yantra Software Solutions (India) Pvt. Ltd.)
"System Generated letter no need of Signature"

ACKNOWLEDGEMENTS

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I express my cordial thanks to **Dr. Duraipandian N**, Dean PSCS & PSIS and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

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Thanush G Gowda

Abstract

Smallholder farmers in developing countries face challenges resulting in reduced farmer incomes and widespread supply chain inefficiencies. The lack of transparent pricing mechanisms, real-time agricultural data, and digital decision-support tools further leaves farmers vulnerable to unpredictable weather, crop diseases, and poor planting decisions. In this project, we design and develop a technology-driven solution to address these structural inequities for market access. We conceptualized and developed a system named Namma Raitha, a multilingual mobile application that eliminates intermediaries by directly connecting farmers with retailers, enabling fair and transparent trade. We implemented listing of agricultural produce with images and quantities, AI-suggested pricing derived from data.gov.in. We integrated a soil-based crop recommendation system using Nitrogen, Phosphorus, Potassium, and pH values. Apart from this, a Convolutional Neural Network (CNN) model for image-based plant disease detection was also implemented that achieved an accuracy of 80.6%. The response time of the system was measured to be under 10 seconds, validating the system's overall efficiency. This project demonstrated that integrating direct market linkage with AI-powered agricultural tools can meaningfully promote transparency, support informed farmer decision-making, and serve as a practical step towards equitable profits and sustainable rural economies.